

Sum and Difference Formulas

ANSWER KEY



Exact values with sum and difference formulas

- 1 Use a sum or difference formula to find the exact value of $\cos 75^\circ$.
 $\cos 75^\circ = \cos(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$.
- 2 Find the exact value of $\sin \frac{\pi}{12}$.
 $\sin \frac{\pi}{12} = \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) = \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{6} - \sqrt{2}}{4}$.
- 3 Find the exact value of $\tan 105^\circ$.
 $\tan 105^\circ = \tan(60^\circ + 45^\circ) = \frac{\sqrt{3} + 1}{1 - \sqrt{3}}$. Rationalizing: $\frac{(\sqrt{3} + 1)^2}{(1 - \sqrt{3})(1 + \sqrt{3})} = \frac{4 + 2\sqrt{3}}{-2} = -2 - \sqrt{3}$.

Working with given values

- 4 Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:
a $\sin(A + B)$ $\cos A = \frac{3}{5}$, $\sin B = \frac{12}{13}$, so $\sin(A + B) = \frac{4}{5} \cdot \frac{5}{13} + \frac{3}{5} \cdot \frac{12}{13} = \frac{56}{65}$.
b $\cos(A + B)$ $\cos(A + B) = \frac{3}{5} \cdot \frac{5}{13} - \frac{4}{5} \cdot \frac{12}{13} = -\frac{33}{65}$.
c the quadrant of $A + B$ Sine positive, cosine negative: Quadrant II.
- 5 Use a sum or difference formula to simplify:
a $\sin \left(x + \frac{\pi}{2} \right) = \cos x$ **b** $\cos(x - \pi) = -\cos x$
- 6 Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A - B)$ exactly.
In QII, $\cos A = -\frac{15}{17}$ and $\cos B = -\frac{4}{5}$.
 $\cos(A - B) = \cos A \cos B + \sin A \sin B = \left(-\frac{15}{17} \right) \left(-\frac{4}{5} \right) + \frac{8}{17} \cdot \frac{3}{5} = \frac{60}{85} + \frac{24}{85} = \frac{84}{85}$.

Verifying identities using sum and difference formulas

- 7 Verify the identity $\cos(x + y) + \cos(x - y) = 2\cos x \cos y$.
 $\cos(x + y) = \cos x \cos y - \sin x \sin y$ and $\cos(x - y) = \cos x \cos y + \sin x \sin y$. Adding: the $\sin x \sin y$ terms cancel, leaving $2\cos x \cos y$.
- 8 Verify the identity $\sin(x + y) - \sin(x - y) = 2\cos x \sin y$.
 $\sin(x + y) = \sin x \cos y + \cos x \sin y$ and $\sin(x - y) = \sin x \cos y - \cos x \sin y$. Subtracting: the $\sin x \cos y$ terms cancel, leaving $2\cos x \sin y$.
- 9 Show that $\tan \left(x + \frac{\pi}{4} \right) = \frac{1 + \tan x}{1 - \tan x}$ using the tangent sum formula $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$.
With $B = \frac{\pi}{4}$, $\tan B = 1$: $\tan \left(x + \frac{\pi}{4} \right) = \frac{\tan x + 1}{1 - \tan x \cdot 1} = \frac{1 + \tan x}{1 - \tan x}$.

Applications

- 10** Two musical tones combine to form a sound wave modeled by $y = \sin\left(x + \frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.

$$\sin\left(x + \frac{\pi}{6}\right) = \sin x \cos \frac{\pi}{6} + \cos x \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x. \text{ The coefficient of } \cos x \text{ is } \frac{1}{2}.$$

Additional practice

- 11** Find the exact value of $\cos \frac{\pi}{12}$.

$$\cos \frac{\pi}{12} = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

- 12** Angles A and B satisfy $\tan A = \frac{3}{4}$ (QI) and $\tan B = \frac{5}{12}$ (QI). Find $\tan(A + B)$ using the tangent sum formula.

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{3}{4} + \frac{5}{12}}{1 - (\frac{3}{4})(\frac{5}{12})} = \frac{\frac{14}{12}}{1 - \frac{15}{48}} = \frac{\frac{7}{6}}{\frac{33}{48}} = \frac{7}{6} \cdot \frac{48}{33} = \frac{56}{33}.$$

- 13** Simplify $\cos\left(\frac{\pi}{2} - x\right)\cos x + \sin\left(\frac{\pi}{2} - x\right)\sin x$ using the cosine difference formula in reverse.

$$\text{This matches the pattern } \cos A \cos B + \sin A \sin B = \cos(A - B) \text{ with } A = \frac{\pi}{2} - x, B = x: = \cos\left(\frac{\pi}{2} - 2x\right).$$

A worked derivation

- 14** Starting from the cosine difference formula $\cos(A - B) = \cos A \cos B + \sin A \sin B$, derive the cosine sum formula $\cos(A + B)$ by substituting $B \rightarrow -B$.

$$\begin{aligned} \cos(A + B) &= \cos(A - (-B)) = \cos A \cos(-B) + \sin A \sin(-B). \text{ Since cosine is even and sine is odd:} \\ &= \cos A \cos B - \sin A \sin B. \end{aligned}$$

- 15** Find the exact value of $\sin 15^\circ \cos 15^\circ$ using a sum/difference-related double angle shortcut (recall $\sin(2\theta) = 2\sin \theta \cos \theta$).

$$\sin 15^\circ \cos 15^\circ = \frac{1}{2}(2\sin 15^\circ \cos 15^\circ) = \frac{1}{2} \sin 30^\circ = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Sum and difference formulas with tables of values

- 16** Given $\sin A = \frac{7}{25}$ (QI) and $\cos B = \frac{4}{5}$ (QI), find $\sin(A - B)$ exactly.

$$\cos A = \frac{24}{25}, \sin B = \frac{3}{5}. \sin(A - B) = \sin A \cos B - \cos A \sin B = \frac{7}{25} \cdot \frac{4}{5} - \frac{24}{25} \cdot \frac{3}{5} = \frac{28}{125} - \frac{72}{125} = -\frac{44}{125}.$$

- 17** Given $\tan A = 2$ (QI) and $\tan B = 3$ (QI), find $\tan(A - B)$ using the tangent difference formula.

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{2 - 3}{1 + 6} = -\frac{1}{7}.$$

Expanding a general sinusoidal expression

- 18 Expand $\cos\left(x - \frac{\pi}{3}\right)$ fully using the difference formula, leaving the answer in terms of $\sin x$ and $\cos x$.
 $\cos\left(x - \frac{\pi}{3}\right) = \cos x \cos \frac{\pi}{3} + \sin x \sin \frac{\pi}{3} = \frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x$.
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A geometric interpretation

- 19 Explain, using the unit circle, why $\cos(A - B)$ (not $\cos A - \cos B$) correctly gives the cosine of the angle between two points at angles A and B on the unit circle.
The angle between two points on the unit circle at angles A and B is $|A - B|$, and by definition the cosine of that INCLUDED angle is $\cos(A - B)$ — a genuinely different quantity from the difference of the individual cosine values $\cos A - \cos B$, which has no direct geometric meaning as an angle's cosine.
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Solving equations using sum and difference formulas

- 20 Solve $\sin\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$ for $x \in [0, 2\pi)$ by first expanding the left side.
Expand: $\sin x \cos \frac{\pi}{6} + \cos x \sin \frac{\pi}{6} = \frac{1}{2} \Rightarrow \frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x = \frac{1}{2}$. Alternatively, recognizing $\sin(\theta) = \frac{1}{2}$ directly: $x + \frac{\pi}{6} = \frac{\pi}{6}$ or $x + \frac{\pi}{6} = \frac{5\pi}{6}$, giving $x = 0$ or $x = \frac{2\pi}{3}$.
- 21 Given that a signal is modeled by $y = \cos\left(3x - \frac{\pi}{4}\right)$, expand it using the difference formula and identify the coefficients of $\cos 3x$ and $\sin 3x$.
 $\cos\left(3x - \frac{\pi}{4}\right) = \cos 3x \cos \frac{\pi}{4} + \sin 3x \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} \cos 3x + \frac{\sqrt{2}}{2} \sin 3x$. Coefficients: both $\frac{\sqrt{2}}{2}$.
- 22 Two AC voltage signals are given by $V_1(t) = \sin(\omega t)$ and $V_2(t) = \sin\left(\omega t + \frac{2\pi}{3}\right)$ (phase-shifted by 120° , common in three-phase power systems). Expand $V_2(t)$ fully using the sum formula, in terms of $\sin(\omega t)$ and $\cos(\omega t)$.
 $V_2(t) = \sin(\omega t) \cos \frac{2\pi}{3} + \cos(\omega t) \sin \frac{2\pi}{3} = -\frac{1}{2} \sin(\omega t) + \frac{\sqrt{3}}{2} \cos(\omega t)$.